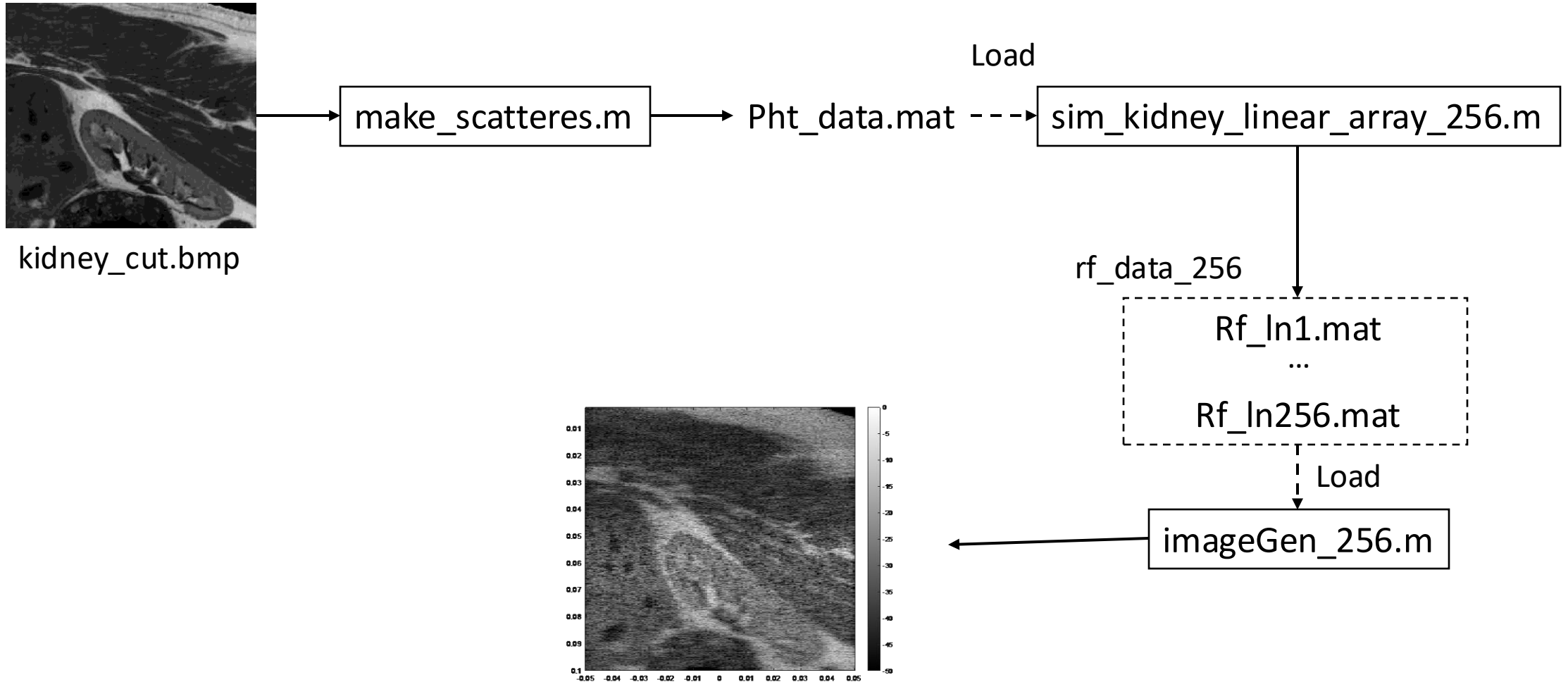


Overview of materials

- Filed_II_mac: Field II libraries for mac
- Field_II_windows: Field II libraries for windows
- **make_scatterers.m**: Generate scatter position/amplitude data of kidney phantom
- human_kidney_phantom.m: Set the scatter parameter of kidney phantom
- bmpread.m: Read bmp data from MRI image
- kidney_cut.bmp: Kidney image
- **sim_kidney_linear_array_256.m**: Generate RF data corresponding to imaging line
- **imageGen_256.m**: Reconstruct the B-mode image with generated RF data
- **DAS.m**: Perform the reconstruction with Delay-and-Sum
- Validation_linear_array.m: validate the DAS.m with point target

Architecture



Generate kidney phantom data

- For making phantom, scatter data (position/amplitude) is required
- `make_scatterers.m`: set the number of scatters
 - Increasing the number e.g. $10e^6$ improves the phantom resolution but increases computational time
- `human_kidney_phantom.m`: define the phantom condition
 - Phantom size, amplitude, coordinate

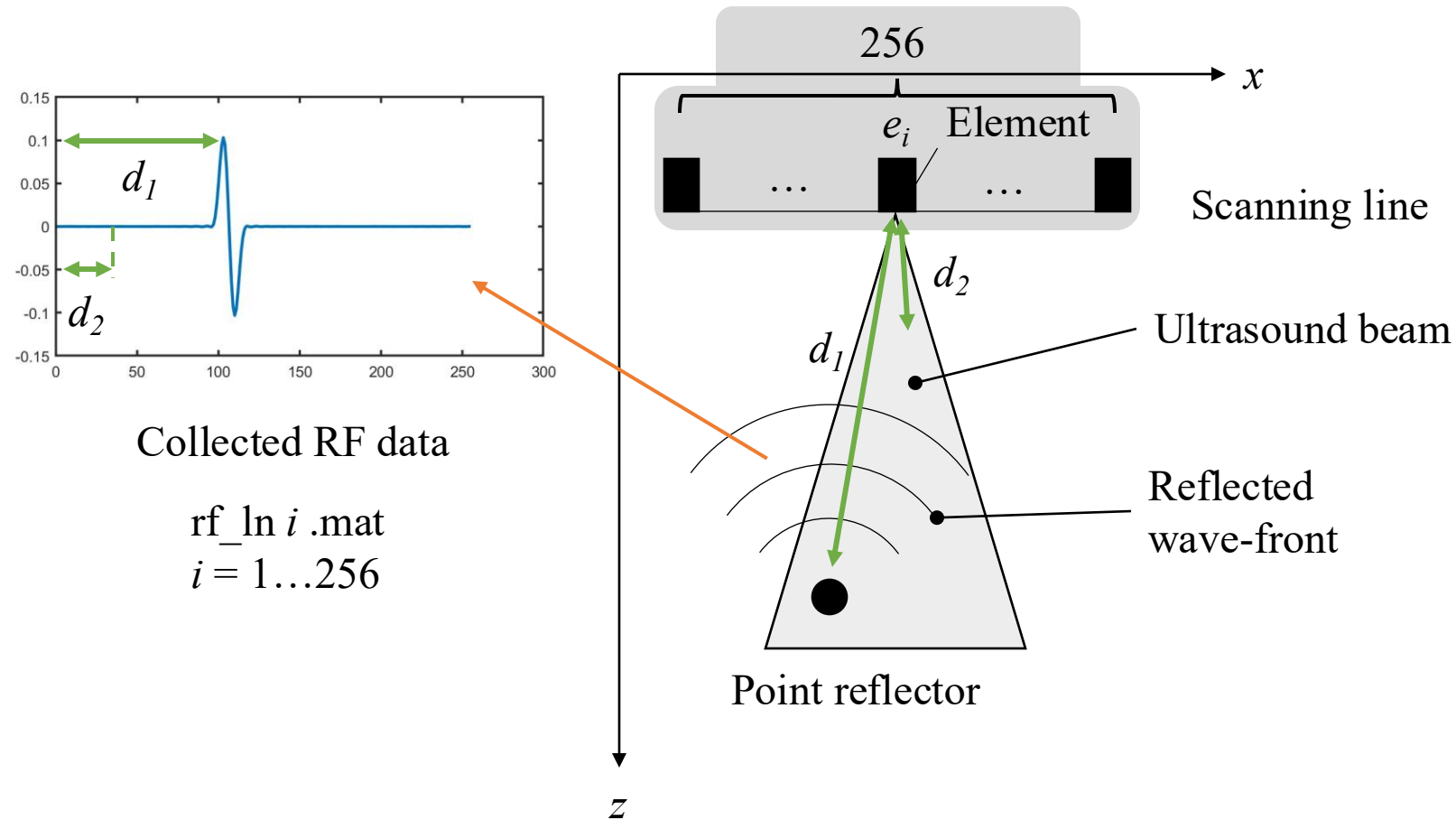
Generate RF data in each element

- For generating RF data, setting configuration of transducer elements are required
- Sim_kidney_linear_array_256.m
 - Set configuration of transducer (Don't need to change it in the homework)
 - Load phantom data
 - Set the number and pitch of transducer elements
 - Calculate RF data in each transducer element
- ✓ This program simply performs to repeat the following process 256 times
 - Transmit/receive the ultrasound beam with single element
 - Move the single element position
 - Transmit/receive the ultrasound beam with single element
 - Move the single element position

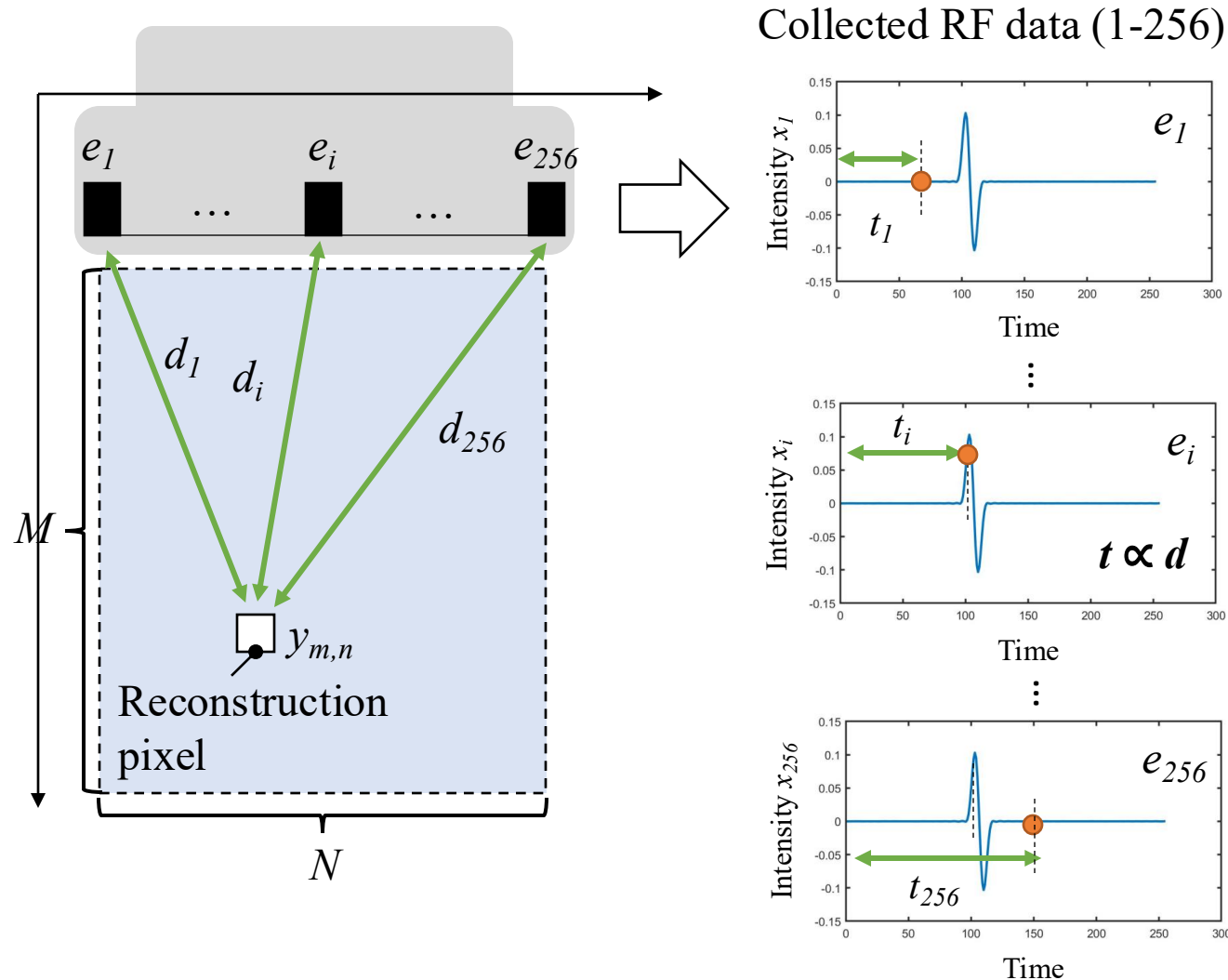
Reconstruct the B-mode image

- To reconstruct B-mode image with collected RF data, beamforming is required
- ImageGen_256
 - Load generated RF data
 - Perform back-projections (delay-and-sum) of generated RF data to the reconstructed plane
 - Convert from projected intensity data to decibel data for visualizing B-mode images
- Delay-and-Sum beamforming
 - Well-known beamforming technique
 - Sum all RF data at the delay that the signal is reflected from the intended reconstruction pixel in each element
 - Detail algorithm here: <https://ieeexplore.ieee.org/document/384453>

Overview of RF data generation process



Delay-and-Sum beamforming

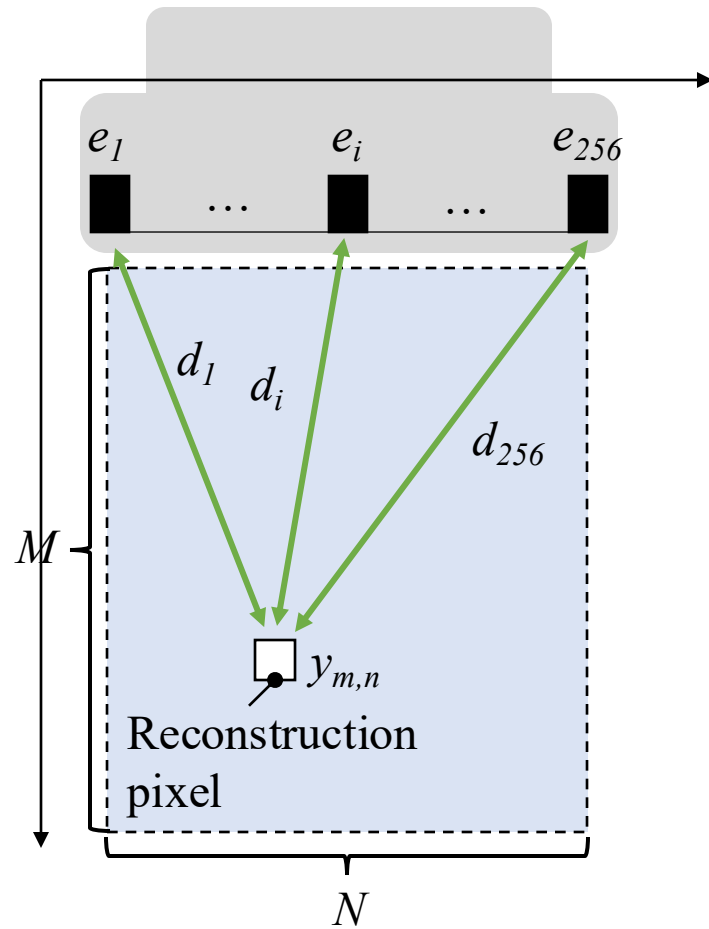


Delay-and-Sum

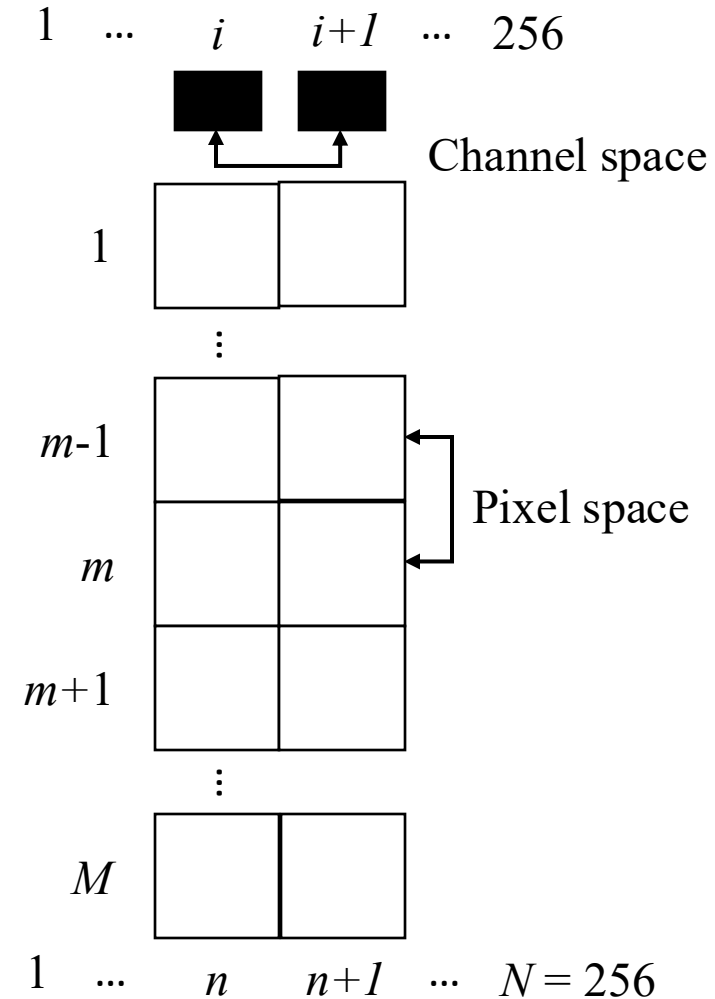
- Sum all RF data at the delay that the signal is reflected from the intended reconstruction pixel in each element
- Need to convert from the distance between the element position and reconstruction pixel to the time
- **Need to calculate the distance**

$$y(m, n) = \sum_{i=1}^{256} x_i(t_i)$$

Delay-and-Sum beamforming



Calculation hint of DAS.m



Validate DAS with point target

- Validate your completed DAS code with “validation_linear_array.m” before testing kidney phantom.
 - Run simply validation_linear_array.m after completing your “DAS.m”
 - You may obtain following image visualizing several point targets if your code is working

